

Lesson 3 Button Control

This section demonstrates controlling the color of RGB lights through different keys on the expansion board.

1. Working Principle

According to the pressed keys, detect the color state of the RGB lights controlled by GPIO port's high and low voltage levels.

```
22 def handle_sigint(signal, frame):
23     global start
24     start = False
25     board.set_rgb([[1, 0, 0, 0], [2, 0, 0, 0]]) # 设置 RGB 为黑色, 关闭所有灯 (
26     # set the RGB to black and turn off all lights)
27 # 注册信号处理程序, 捕获 Ctrl+C 信号 (register a signal process program to captu
28 # e the Ctrl+C signal)
29 signal.signal(signal.SIGINT, handle_sigint)
30 try:
31     key1_pin = 13
32     chip = gpiod.Chip("gpiochip4")
33     key1 = chip.get_line(key1_pin)
34     key1.request(consumer="key1", type=gpiod.LINE_REQ_DIR_IN, flags=gpiod.LINE_
35     EQ_FLAG_BIAS_PULL_UP)
36     key2_pin = 23
37     key2 = chip.get_line(key2_pin)
38     key2.request(consumer="key2", type=gpiod.LINE_REQ_DIR_IN, flags=gpiod.LINE_
39     EQ_FLAG_BIAS_PULL_UP)
40     while start:
41         # 检查按键状态 (check button status)
42         key1_state = key1.get_value()
43         key2_state = key2.get_value()
44
45         if key1_state == 0:
46             # 按键1被按下, 设置 RGB 为红色 (press key1 to set RGB to red)
47             board.set_rgb([[1, 255, 0, 0], [2, 255, 0, 0]])
48
```

① Initialize the Raspberry Pi expansion board and GPIO pins, handle the "Ctrl+C" signal, and set a signal handling function to exit the program. In a loop, constantly check the status of two buttons.

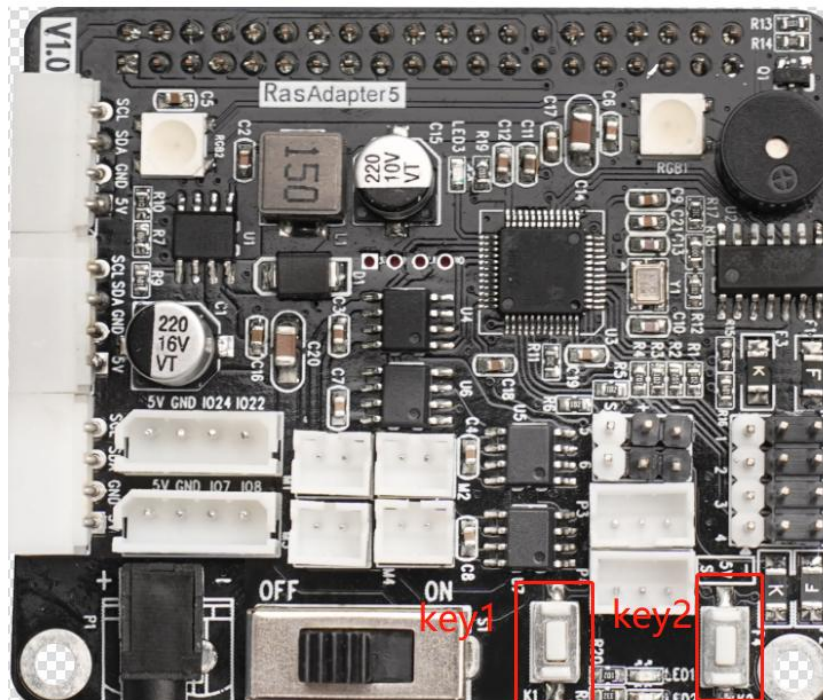
② If key1 is pressed, set the color of RGB lights to red.

- ③ If key2 is pressed, set the color of RGB light to blue.
- ④ Print the key state. (press or release)
- ⑤ Perform in a loop until receive the Ctrl+C signal to end the program.
- ⑥ The source code of this program is located in:

/home/pi/board_demo/key.py

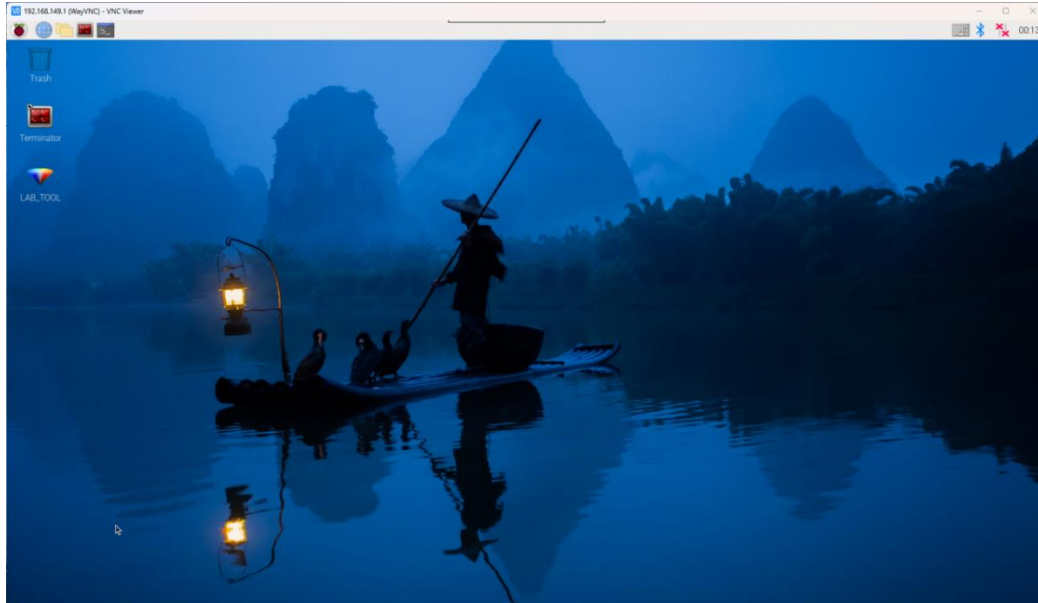
2. Getting Ready

The Raspberry Pi expansion board has two keys as pictured:

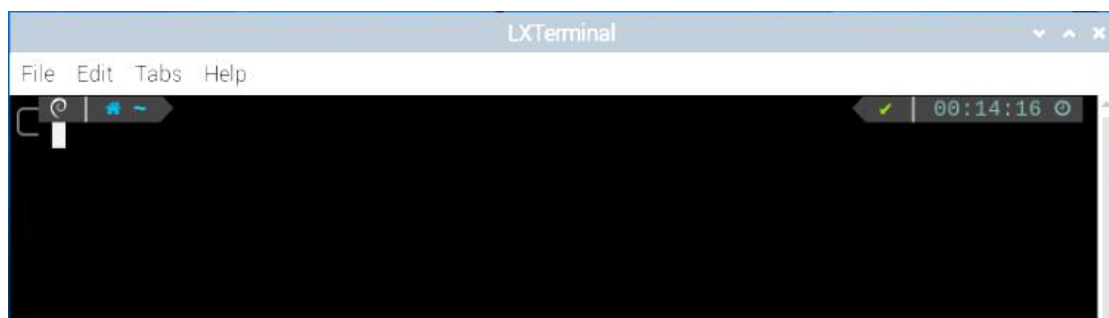


3. Operation Steps

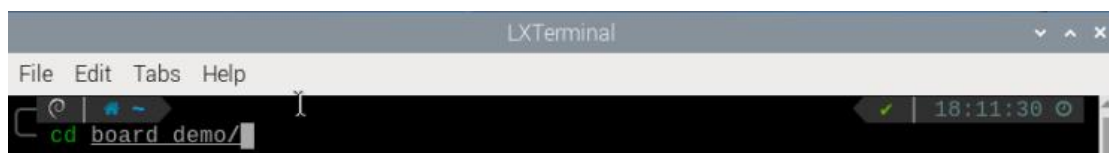
- 1) Turn the device on and connect the robot through the VNC remote connection tool.



- 2) Click the icon  in the left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



- 3) Enter the command “cd board_demo/” in the directory where the program is located.



- 4) Next, enter the command “python3 rgb_control_demo.py” to run the program.

```
LXTerminal
File Edit Tabs Help
> cd board_demo
python3 key.py

*****功能:幻尔科技树莓派扩展板, 按键控制RGB灯*****
*****
-----
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
-----
Tips:
* 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试!
-----

发生异常: [Errno 16] Device or resource busy
按键默认被 hw_button_scan 占用, 需要先关闭服务
sudo systemctl stop hw_button_scan.service
```

5) Input command “sudo systemctl stop hw_button_scan.service”.

```
> sudo systemctl stop hw_button_scan.service
~/.board_demo 00:15:51
```

6) Input command “python3 key.py”. Then press enter key. And it will print a prompt.

```
pi@raspberrypi:~/board_demo $ python3 key.py

*****功能:幻尔科技树莓派扩展板, 按键控制RGB灯(function:Hiwonder Raspberry Pi expansion board, button control for RGB LED)*****
*****
-----
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
-----
Tips:
* 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试! (Press "Ctrl+C" to exit the program. If it fails, please try again multiple times!)
-----

key1: 1 key2: 1
```

Press key1 to set the light to red. Press key2 to set the light to blue. Press CTRL+C to close the program.

7) After completing the program, it is necessary to restart the “hw_button_scan” service. Input the command to restart the service command “sudo systemctl start hw_button_scan.service”.

```
sudo systemctl start hw_button_scan.service
```

4. Project Outcome

- If key1 is pressed, use the “board.set_rgb()” function to set the RGB color to red.
- If key2 is pressed, use the “board.set_rgb()” function to set the RGB color to blue.
- Print the statues of the keys (key1_state and key2_state) on the console. Then press CTRL+C to exit the procedure.